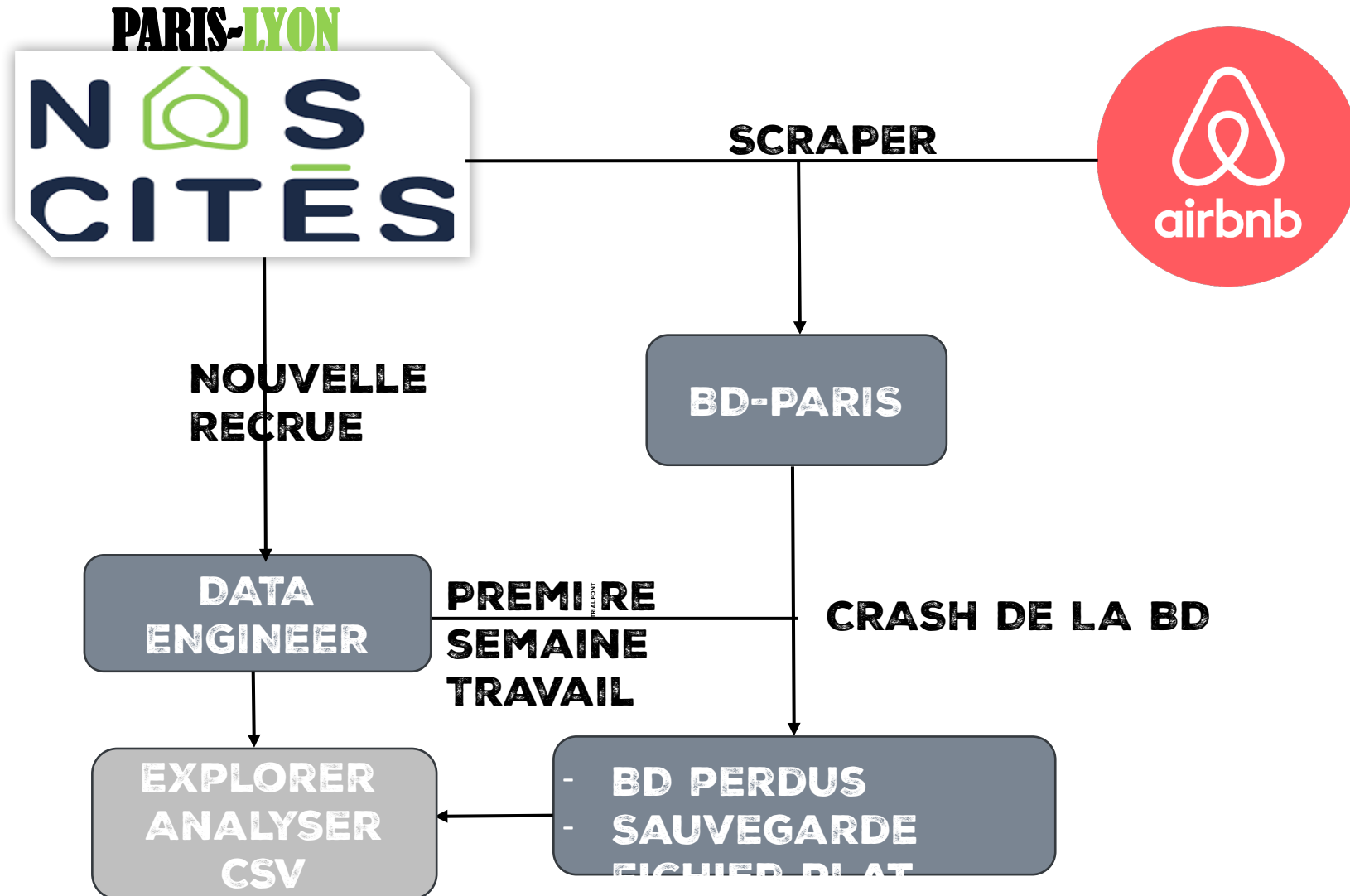


CONCEVER ET ANALYSE UNE BASE DE DONNEES MONGODB

CONTEXTE



PLAN DE PRESENTATION



LES BENEFICES DU NOSQL

IMPORTER LES
DONNÉES
COMPRENDRE LES



ANALYSE DE LA BD MONGODB

RECHERCHER LES DONNÉES
POLAIRES ET REQUÊTES
COMPLEXES
CONNECTER LA BD



CONCEPTION DE LA BD MONGODB

REPLIQUER LES
DONNÉES
DISTRIBUER LES

```
Microsoft Windows [version 10.0.26100.6584]
(c) Microsoft Corporation. Tous droits réservés.

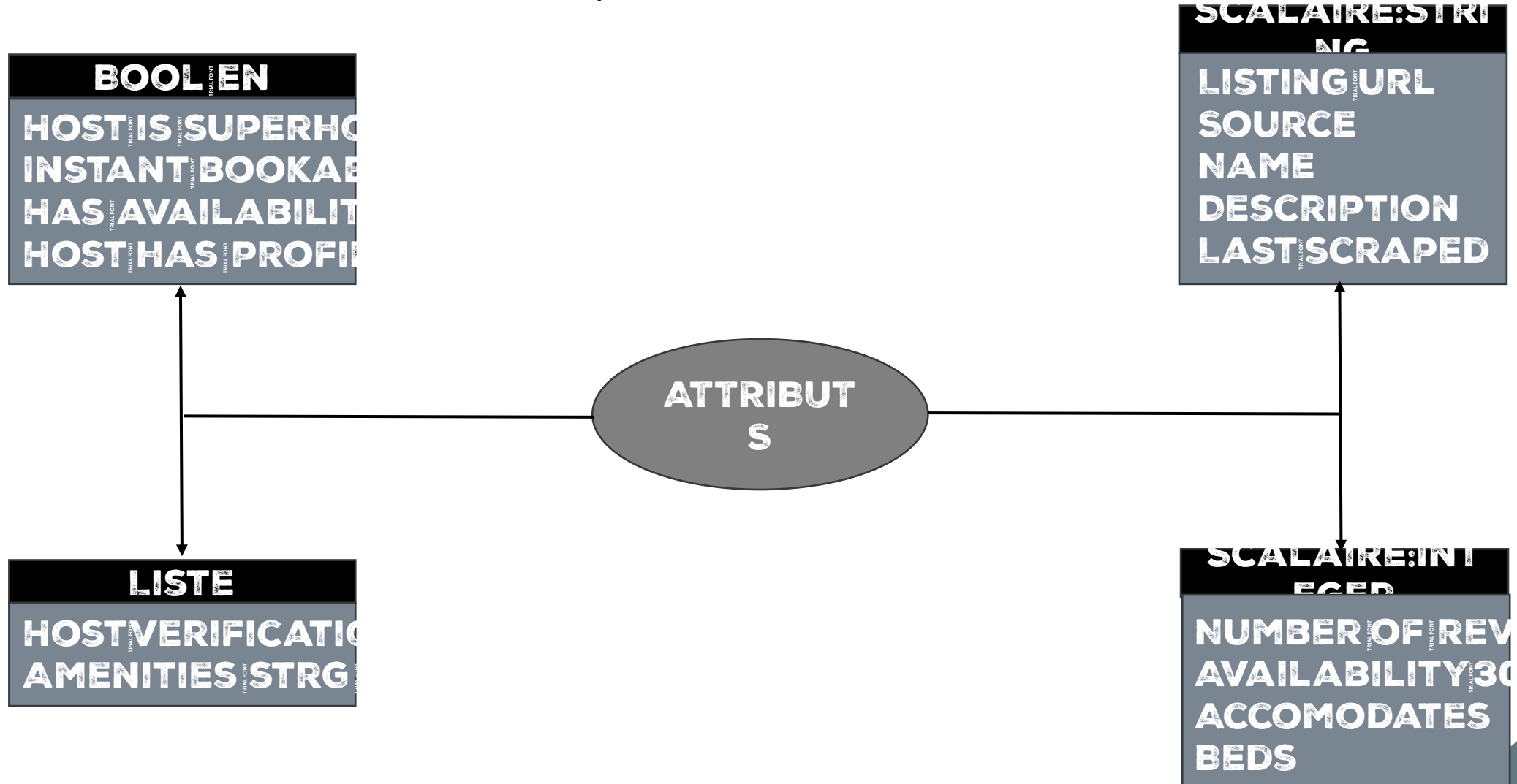
C:\Users\emako>cd C:\Mongo_tools\mongodb-database-tools-windows-x86_64-100.12.2\mongodb-database-tools-windows-x86_64-100.12.2\bin

C:\Mongo_tools\mongodb-database-tools-windows-x86_64-100.12.2\mongodb-database-tools-windows-x86_64-100.12.2\bin>mongoimport --db logements_paris --collection logements --type csv --headerline --file C:\Users\emako\Desktop\Open_projet_VII\listings_Paris.csv
2025-09-24T11:09:43.093+0200    connected to: mongodb://localhost/
2025-09-24T11:09:46.093+0200    [#####.....] logements_paris.logements    49.7MB/185MB (26.9%)
2025-09-24T11:09:49.093+0200    [#####.....] logements_paris.logements    72.7MB/185MB (39.4%)
2025-09-24T11:09:52.093+0200    [#####.....] logements_paris.logements    97.1MB/185MB (52.6%)
2025-09-24T11:09:55.093+0200    [#####.....] logements_paris.logements    121MB/185MB (65.3%)
2025-09-24T11:09:58.093+0200    [#####.....] logements_paris.logements    144MB/185MB (77.9%)
2025-09-24T11:10:01.093+0200    [#####.....] logements_paris.logements    167MB/185MB (90.7%)
2025-09-24T11:10:03.442+0200    [#####.....] logements_paris.logements    185MB/185MB (100.0%)
2025-09-24T11:10:03.442+0200    95885 document(s) imported successfully. 0 document(s) failed to import.

C:\Mongo_tools\mongodb-database-tools-windows-x86_64-100.12.2\mongodb-database-tools-windows-x86_64-100.12.2\bin>
```

LES BÉNÉFICES DU NOSQL: DICTIONNAIRE DE DONNÉES

TAILLES : 95 885 ENREGISTREMENTS, 76 ATTRIBUTS



LES BENEFICES DU NOSQL: AVANTAGES DE MONGODB

- FLEXIBILITE DU SCHEMA : HOST ABOUT, NEIGHBORHOOD OVERVIEW, AMENITIES, ETC.....
- VOLUME DE DONNEES : 1M, SCALABILITE HORIZONTALE COMPLEXE EN SQL.
- GESTION NATIVES DES DONNEES COMPLEXE : LISTES
- PERFORMANCE POUR LECTURE ET ECRITURE MASSIVE : LES SCRAPERS INJECTENT LES MILLIERS D'ANONCES RAPIDEMENT.
- MISE A JOUR DES DOCUMENTS SANS GESTION DE PLUSIEURS TABLES LIEES.
- LATENCE FAIBLE.

ANALYSE DE LA BD MONGODB: REQUETES SIMPLES ET RESULTATS

- **NOM DE LA BD: LOGEMENTS PARIS**
- **NOMBRE DE COLLECTIONS: 01 NOMME LOGEMENTS**
- **NOMBRE DE DOCUMENTS DANS LA COLLECTION:**
DB.GETCOLLECTION LOGEMENTS.FIND().COUNT R P 95 885
- **NOMBRE DE LOGEMENTS AVEC DISPONIBILITES:**
DB.GETCOLLECTION LOGEMENTS.FIND(HAS AVAILABILITY: T).COUNT R P: 90173

ANALYSE DE LA BD MONGODB: REQUETES SIMPLES ET RESULTATS

COMBIEN D'ANNONCES Y A-T-IL PAR TYPE DE LOCATION?

```
VARGROUP GROUP: ID:: PROPERTY TYPE, TOTAL: SUM:1
VARDECROI SORT: TOTAL:-1
DB.GETCOLLECTION LOGEMENTS .AGGREGATE VARGROUP,
```

_id	total	_id	total
Entire rental unit	80516.0	Shared room in boutique hotel	3.0
Private room in rental unit	5980.0	Shared room in home	3.0
Entire condo	2679.0	Private room in vacation home	3.0
Room in boutique hotel	1256.0	Shared room in townhouse	3.0
Entire loft	1007.0	Cave	2.0
Room in hotel	1003.0	Shared room in tiny home	2.0
Entire home	696.0	Room in rental unit	2.0
Private room in bed and breakfast	590.0	Dome	1.0
Private room in condo	344.0	Shared room	1.0
Entire serviced apartment	277.0	Shared room in farm stay	1.0
Shared room in rental unit	274.0	Private room in earthen home	1.0
Entire townhouse	268.0	Entire bed and breakfast	1.0
Private room in home	122.0	Shared room in cabin	1.0
Private room in townhouse	86.0	Shipping container	1.0
Private room in loft	73.0	Shared room in serviced apartment	1.0
Room in aparthotel	73.0	Tower	1.0
Shared room in hostel	54.0	Barn	1.0
Private room in hostel	53.0	Private room in cave	1.0
Entire place	46.0	Castle	1.0
Private room in guesthouse	45.0	Shared room in guest suite	1.0
Private room in guest suite	38.0	Shared room in ice dome	1.0
Entire vacation home	37.0	Entire bungalow	1.0
Private room	37.0	Private room in villa	1.0

ANALYSE DE LA BD MONGODB: REQUÊTES SIMPLES ET RÉSULTATS

QUELLES SONT LES
5 ANNONCES DE LOCATION
AVEC LE
PLUS D'ÉVALUATIONS ?
ET COMBIEN D'ÉVALUATIONS
ONT-ELLES ?

The screenshot shows the MongoDB Shell interface with the following components:

- Tabs:** Quickstart, IntelliShell: Local MongoDB*, IntelliShell: Local MongoDB*.
- Path:** localhost:27017 > logements_paris
- Toolbar:** Play, Stop, IntelliShell icon, New, Load file, Save file, Enable Query Assist, Change view.
- Code Editor:**

```
1 Varpro={ $project: { "_id": 1, "number_of_reviews": 1}}
2 Varsort={$sort: { "number_of_reviews": -1 } }
3 db.getCollection("logements").aggregate([Varpro, Varsort, {$limit: 5 }])
4
5
6
```
- Output Tabs:** Raw shell output, Shell Output (Documents), Aggregate Query (line 3).
- Query Execution:** Refresh, Previous, Next, Limit (50), Documents 1 to 5.
- Results Table:**

logements	>	number_of_reviews
<code>_id</code>		<code>number_of_reviews</code>
<code>id</code> 68d3b558bf3515e2dd7cdba5		3067
<code>id</code> 68d3b559bf3515e2dd7cf47f		2620
<code>id</code> 68d3b55abf3515e2dd7d191f		2294
<code>id</code> 68d3b55abf3515e2dd7d1707		2105
<code>id</code> 68d3b559bf3515e2dd7d087e		2048

ANALYSE DE LA BD MONGODB: REQUÊTES SIMPLES ET RESULTATS

QUEL EST LE NOMBRE TOTAL
D'HÔTES DIFFÉRENTS ?

The screenshot shows the MongoDB Studio 3T Community Edition interface. The top bar has tabs for 'Quickstart', 'IntelliShell: Local MongoDB*', and 'IntelliShell: Local MongoDB*'. The main window displays the 'localhost:27017' database with the 'logements_paris' collection selected. The query editor contains the following query:

```
1 db.logements.distinct("host_id").length
```

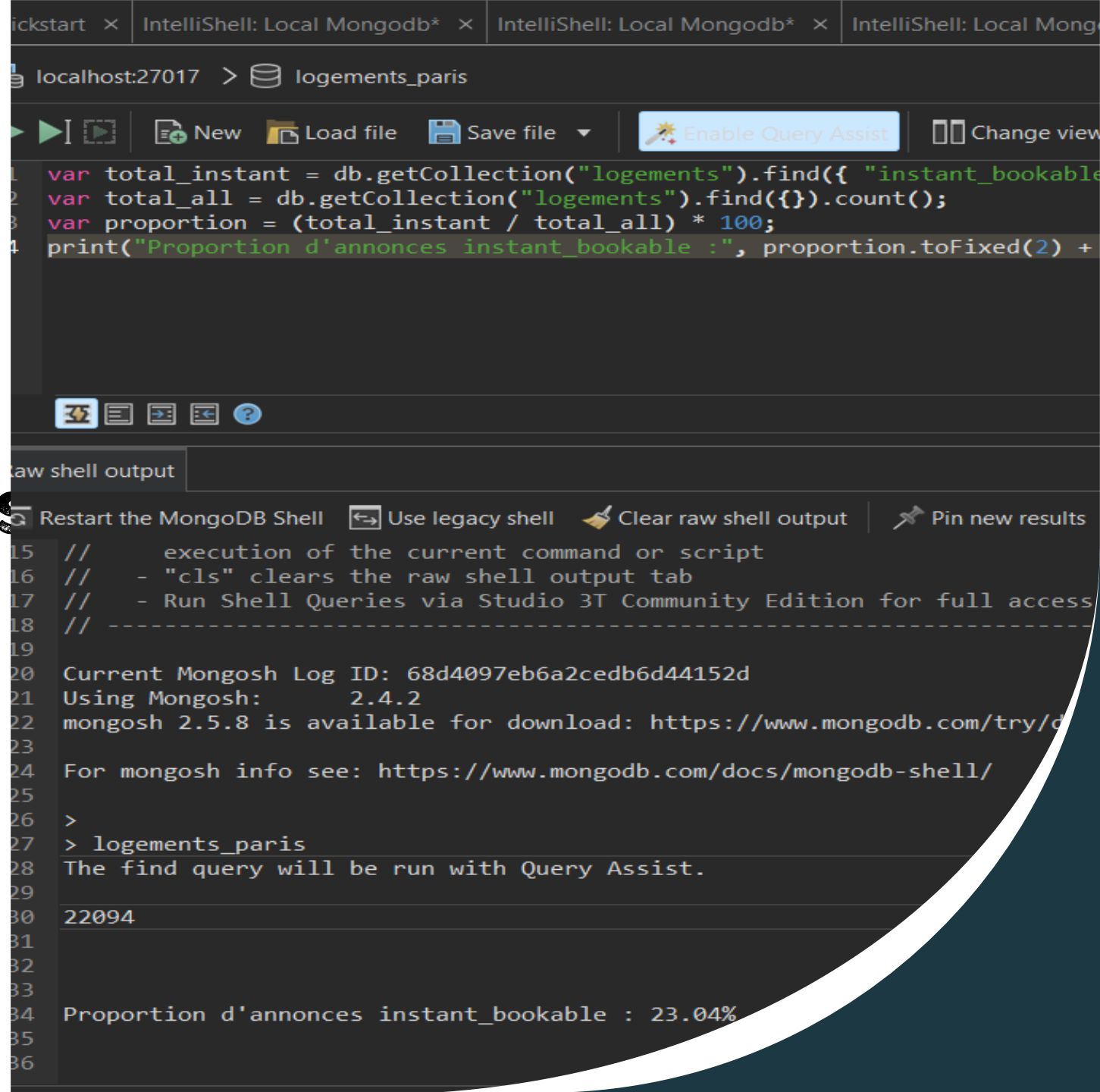
The 'Raw shell output' tab is active, showing the following text:

```
Restart the MongoDB Shell Use legacy shell Clear raw shell output Pin new results

8 // - F6 runs the statement at the current cursor position
9 // - F9 runs the current selection
10 // - Shift+Enter run the current selection, or the entire script
11 // if nothing is selected
12 // - Ctrl+Enter runs the current selection, or the statement at the
13 // current cursor position if nothing is selected
14 // - ESC offers you to restart the shell should you want to cancel
15 // execution of the current command or script
16 // - "cls" clears the raw shell output tab
17 // - Run Shell Queries via Studio 3T Community Edition for full
18 // -----
19
20 Current Mongosh Log ID: 68d408345ef1f6fd0844152d
21 Using Mongosh: 2.4.2
22 mongosh 2.5.8 is available for download: https://www.mongodb.com/
23
24 For mongosh info see: https://www.mongodb.com/docs/mongosh
25
26 >
27 > logements_paris
28 71979
29
```

ANALYSE DE LA BD MONGODB: REQUÊTES SIMPLES ET RESULTATS

Quel est le nombre de locations réservables instantanément ? Cela représente quelle proportion des annonces ?



The screenshot shows the MongoDB Studio 3T interface. The top bar indicates the current database is 'logements_paris' on 'localhost:27017'. The main editor contains a JavaScript query to calculate the proportion of instant bookable listings. The 'Raw shell output' tab is active, displaying the execution of the query and the resulting output.

```
var total_instant = db.getCollection("logements").find({ "instant_bookable": true }).count();
var total_all = db.getCollection("logements").find({}).count();
var proportion = (total_instant / total_all) * 100;
print("Proportion d'annonces instant_bookable :", proportion.toFixed(2) + "%");
```

Raw shell output

```
Restart the MongoDB Shell Use legacy shell Clear raw shell output Pin new results
// execution of the current command or script
// - "cls" clears the raw shell output tab
// - Run Shell Queries via Studio 3T Community Edition for full access
// -----
Current Mongosh Log ID: 68d4097eb6a2cedb6d44152d
Using Mongosh: 2.4.2
mongosh 2.5.8 is available for download: https://www.mongodb.com/try/download/mongosh
For mongosh info see: https://www.mongodb.com/docs/mongodb-shell/
>
> logements_paris
The find query will be run with Query Assist.
22094

Proportion d'annonces instant_bookable : 23.04%
```

ANALYSE DE LA BD MONGODB: REQUÊTES SIMPLES ET RESULTATS

Est-ce que des hôtes ont plus de 100 annonces sur les plateformes ? Et si oui qui sont-ils ? Cela représente quel pourcentage des hôtes ?

The screenshot shows the MongoDB Compass interface. At the top, there are tabs for 'Quickstart' and several 'IntelliShell: Local Mongod...' windows. The main window displays the 'logements_paris' collection in the 'localhost:27017' database. A query is entered in the command bar:

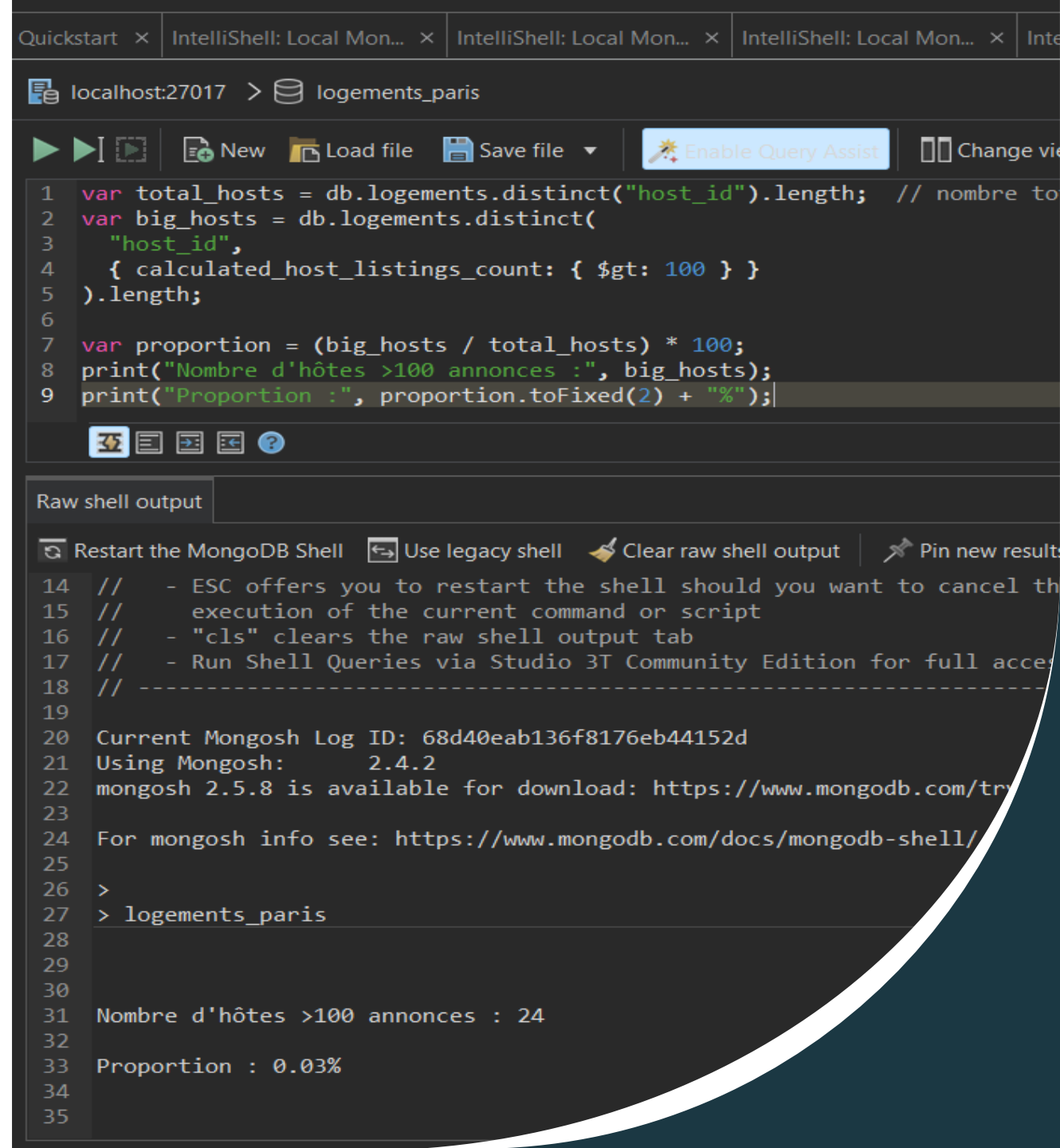
```
1 db.logements.distinct(  
2   "host_id",  
3   { calculated_host_listings_count: { $gt: 100 } }  
4 )
```

Below the query bar, there are icons for 'Raw shell output', 'Shell Output (Array)', and 'Table View'. The 'Table View' is selected, showing a table with 13 documents. The first document is highlighted in blue.

[Index]	
0	1112584.0
1	7642792.0
2	21630783.0
3	24495283.0
4	26981054.0
5	28313443.0
6	33889201.0
7	50502817.0
8	50978178.0
9	51567288.0
10	99040006.0
11	125797498.0
12	314994947.0
13	335998296.0

ANALYSE DE LA BD MONGODB: REQUÊTES SIMPLES ET RESULTATS

Est-ce que des hôtes ont plus de 100 annonces sur les plateformes ? Et si oui qui sont-ils ? Cela représente quel pourcentage des hôtes ?



The screenshot shows the MongoDB Studio interface. The top bar includes tabs for 'Quickstart', 'IntelliShell: Local Mon...', and 'IntelliShell: Local Mon...'. The main editor displays a JavaScript query for the 'logements_paris' database. The query calculates the total number of unique hosts and the number of hosts with more than 100 listings, then calculates the proportion of such hosts.

```
1 var total_hosts = db.logements.distinct("host_id").length; // nombre total
2 var big_hosts = db.logements.distinct(
3   "host_id",
4   { calculated_host_listings_count: { $gt: 100 } }
5 ).length;
6
7 var proportion = (big_hosts / total_hosts) * 100;
8 print("Nombre d'hôtes >100 annonces :", big_hosts);
9 print("Proportion :", proportion.toFixed(2) + "%");
```

The bottom panel shows the 'Raw shell output' with the following text:

```
14 // - ESC offers you to restart the shell should you want to cancel the
15 //   execution of the current command or script
16 // - "cls" clears the raw shell output tab
17 // - Run Shell Queries via Studio 3T Community Edition for full access
18 // -----
19
20 Current Mongosh Log ID: 68d40eab136f8176eb44152d
21 Using Mongosh: 2.4.2
22 mongosh 2.5.8 is available for download: https://www.mongodb.com/tr
23
24 For mongosh info see: https://www.mongodb.com/docs/mongodb-shell/
25
26 >
27 > logements_paris
28
29
30
31 Nombre d'hôtes >100 annonces : 24
32
33 Proportion : 0.03%
34
35
```

ANALYSE DE LA BD MONGODB: REQUÊTES SIMPLES ET RESULTATS

Combien y a-t-il de super hôtes différents ?
Cela représente quel pourcentage des
hôtes ?

The screenshot shows the MongoDB Studio interface. The top bar includes tabs for 'Quickstart' and several 'IntelliShell: Loc...' instances. The main editor displays a JavaScript query to calculate the number and proportion of superhosts. The 'Raw shell output' pane at the bottom shows the execution results, including a warning about the shell version and the final data output.

```
localhost:27017 > logements_paris
```

```
1 var total_hosts = db.logements.distinct("host_id").length
2 var superhosts = db.logements.distinct(
3   "host_id",
4   { host_is_superhost: "t" }
5 ).length;
6 var proportion = (superhosts / total_hosts) * 100;
7 print("Nombre de superhosts :", superhosts);
8 print("Proportion de superhosts :", proportion.toFixed(2) + "%");
```

```
Raw shell output
```

```
Restart the MongoDB Shell Use legacy shell Clear raw shell output Pin new results
14 // - ESC offers you to restart the shell should you want to cancel the
15 //   execution of the current command or script
16 // - "cls" clears the raw shell output tab
17 // - Run Shell Queries via Studio 3T Community Edition for full access
18 // -----
19
20 Current Mongosh Log ID: 68d40fc3377122dd7344152d
21 Using Mongosh: 2.4.2
22 mongosh 2.5.8 is available for download: https://www.mongodb.com/try
23
24 For mongosh info see: https://www.mongodb.com/docs/mongodb-shell/
25
26 >
27 > logements_paris
28
29
30
31 Nombre de superhosts : 10027
32
33 Proportion de superhosts : 13.93%
34
35
```


ANALYSE DE LA BD MONGODB: REQUÊTES COMPLEXES ET RÉSULTATS

Calculer le taux de
réservation moyen par mois
par type de logement

Shape: (70, 2)

property_type --- str	taux_reservation_moyen_ETE_par... --- f64
Private room in bed and breakf...	0.532976
Floor	0.211111
Entire vacation home	0.437838
Private room in boat	0.587654
Private room in tiny home	0.64
...	...
Entire bed and breakfast	0.0
Shared room in hostel	0.238066
Room in boutique hotel	0.543719
Private room in hostel	0.368344
Tower	0.011111

ANALYSE DE LA BD MONGODB: REQUETES COMPLEXES ET RESULTATS

CALCULER LA MEDIANE DES NOMBRE D'AVIS POUR TOUS LES LOGEMENTS
MEDIANE DU NOMBRE D'AVIS : 3.0

CALCULER LA MEDIANE DES NOMBRE D'AVIS PAR CATEGORIE D'HTE

shape: (4, 3)

host_is_superhost	host_identity_verified	median_reviews
---	---	---
str	str	f64
f	f	1.0
t	t	25.0
t	f	15.0
f	t	2.0

ANALYSE DE LA BD MONGODB: REQUETES COMPLEXES ET RESULTATS

CALCULER LA DENSITE DE LOGEMENTS PAR QUARTIER DE PARIS

Nombre de logements par quartier :
shape: (20, 2)

neighbourhood_cleansed ---	nb_logements ---
str	u32
Buttes-Montmartre	10555
Popincourt	8430
Vaugirard	7802
Batignolles-Monceau	6857
Entrepôt	6558
...	...
Élysée	2898
Hôtel-de-Ville	2821
Palais-Bourbon	2740
Luxembourg	2701
Louvre	2026

Total de logements dans tous les quartiers : 95885

ANALYSE DE LA BD MONGODB: REQUÊTES COMPLEXES ET RÉSULTATS

Identifier les quartiers avec le plus fort
taux de réservation par mois

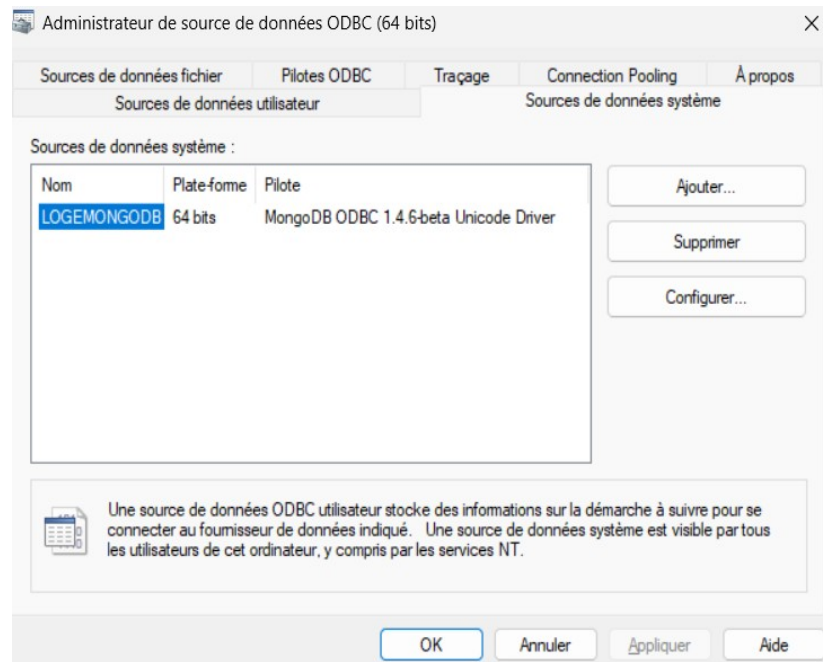
shape: (20, 2)

neighbourhood_cleansed ---	taux_reservation_moyen_quartie... ---
str	f64
Ménilmontant	0.614212
Buttes-Chaumont	0.60267
Popincourt	0.602589
Entrepôt	0.602174
Buttes-Montmartre	0.597726
...	...
Palais-Bourbon	0.538775
Bourse	0.532006
Luxembourg	0.514176
Élysée	0.487973
Passy	0.474011

PROCESSUS DE CONNEXION

1. **Télécharger et installer les logiciels suivants: Connector BI, Visual C, ODBC**
2. **Configurer une source de données ODBC DSN**
→
3. **Lancer Power BI → Cliquer sur Obtenir les données**
Choisir ODBC et cliquer sur se connecter la source ODBC
Choisir la DSN Créer la BD et ensuite la collection et télécharger.

CAPTURE D'ECRAN DES TAPES D'INSERTION



Obtenir les données

odbc

Tout

Base de données

Autre

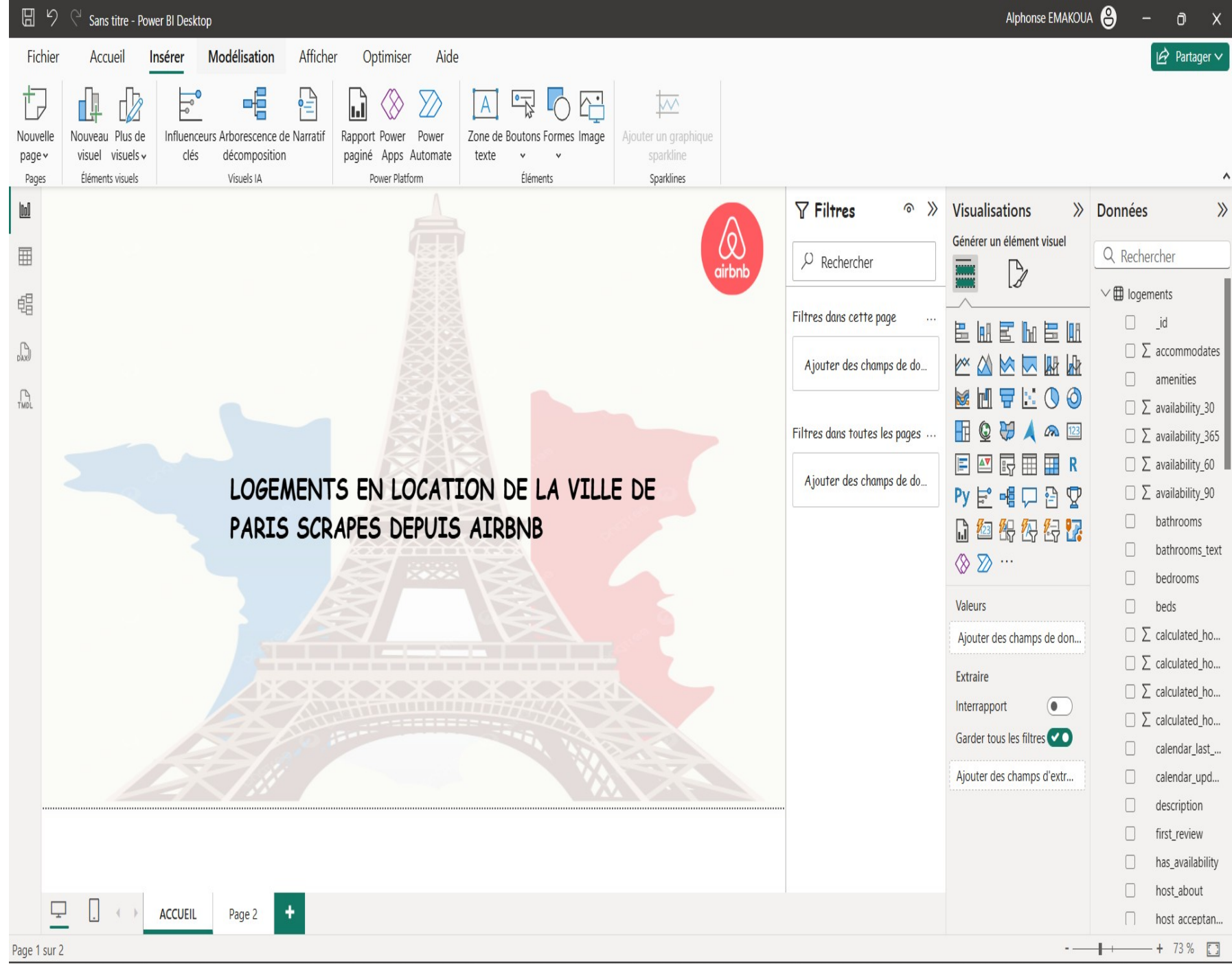
Tout

ODBC

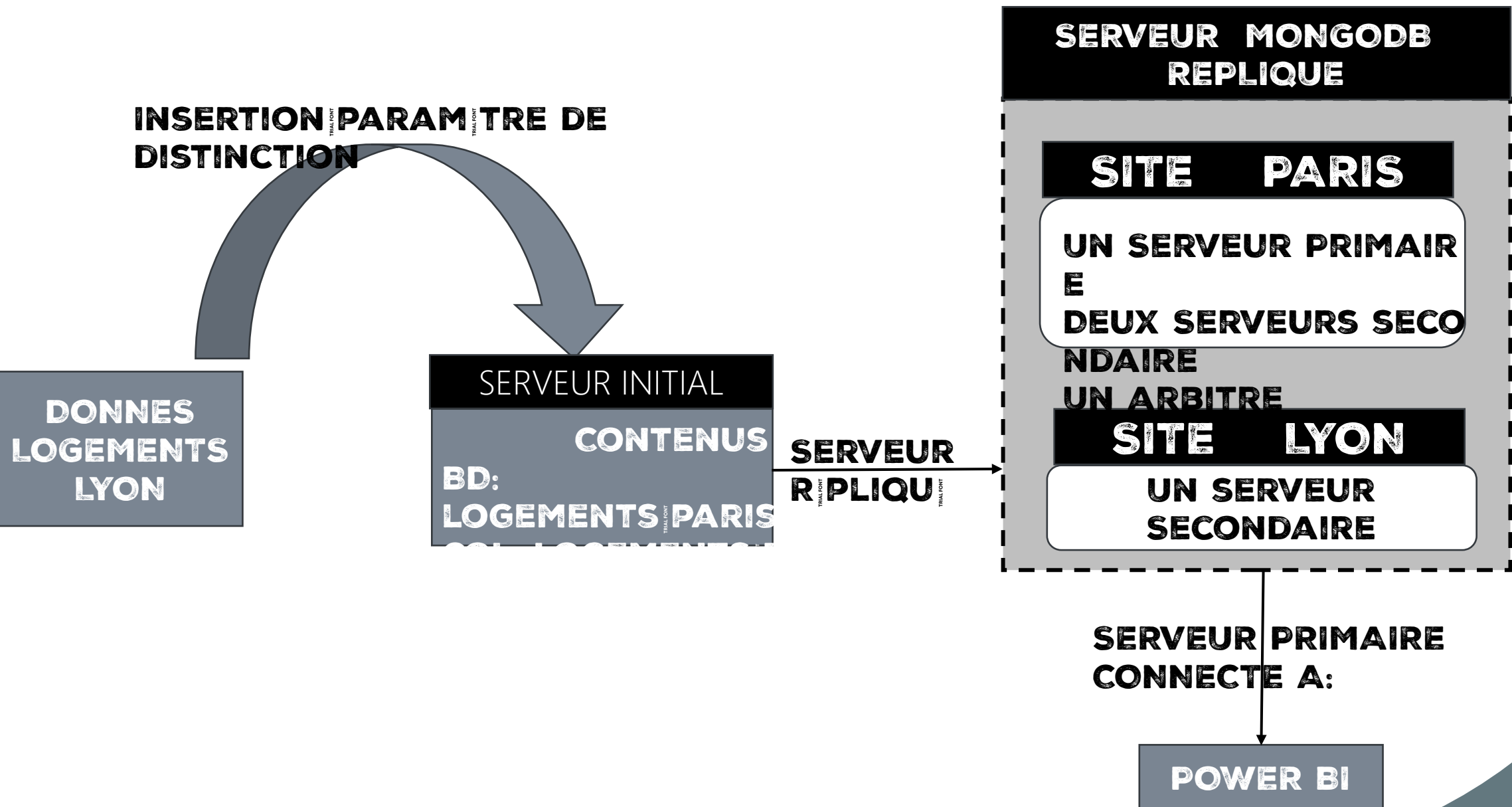
Kyvas ODBC

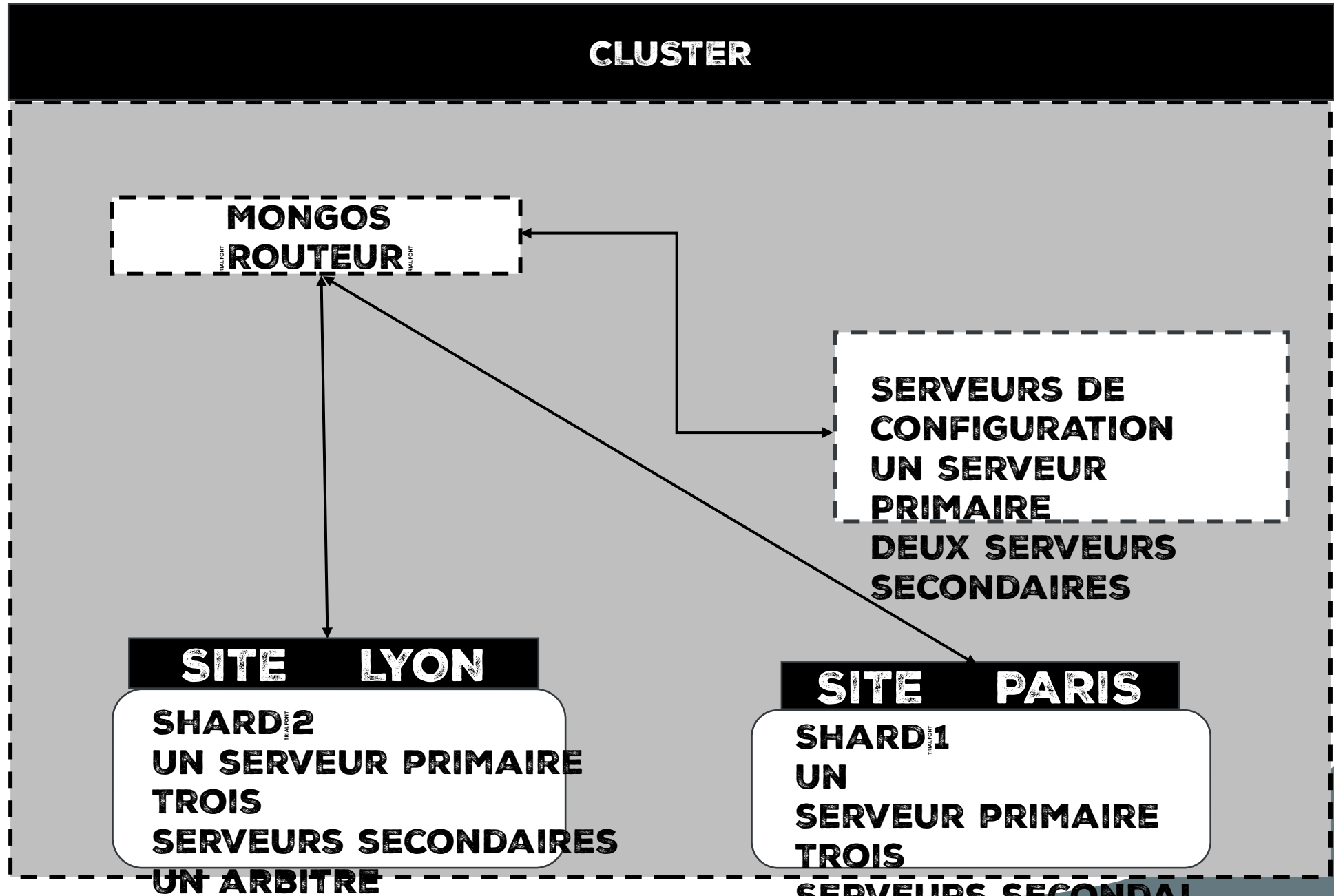
ANALYSE DE LA BD MONGODB: CONNEXION DE LA BD POWER BI

CAPTURE D'ECRAN DU
TABLEAU DE BORD PAGE
D'ACCUEIL PERSONNALISEE



REPLIQUER LES DONNÉES





**MERCI POUR VOTRE
AIMABLE
ATTENTION**

